



CPSC 100

Computational Thinking

Computational Thinking and Markdown

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Agenda

- Welcome New Students!
- Logistics
 - Information about Tests
 - Book your first Test!
- Intro to Computational Thinking (CT)
 - CT Skills
 - Class Activity

Warning:

**Course waitlist will close EOD
of Friday!**

Welcome New Students!

About Tests in this course

Week	Week of	Topic	Test (Content)
1	Jan 5	Introductions	-
2	Jan 12	Algorithms I	T1 (Intro)
3	Jan 19	Algorithms II	T2 (Alg I)
4	Jan 26	Data Representation	T3 (Alg II)
5	Feb 2	Programming I	T4 (Data Rep)
6	Feb 9	Programming II	T5 (Programming I)
7	Feb 16	Reading week	-
8	Feb 23	Internet and HCI	T6 (Programming II)
9	Mar 2	Artificial Intelligence I	T7 (Internet & HCI)
10	Mar 9	Artificial Intelligence II	T8 (AI I)
11	Mar 16	Catch-up week	-
12	Mar 23	Data Mining	T9 (AI II)
13	Mar 30	Data Visualization	T10 (Data mining)
14	Apr 6	Final Projects	T11 (Data Visualization)



About Tests in this course

- Tests will be happening in the Computer-Based Testing Facility (CBTF): cbtf.ubc.ca
- Students have access to class slides and reference material during their Test - **memorization is not necessary!**
 - If you want to request specific material (for the whole class) be included, please request it on Ed Discussion!
- Tests will generally be cumulative, though there will be a focus on new content (less than 25% will be on 'old' content)



About Tests in this course

- Tests will be 50 mins long
- We will drop your lowest Test score
 - We will count your best 10 scores out of 11
- Take the **first one** seriously! It's super easy!
- Tests can be booked from Saturday to Wednesday
- **Reservations will usually become available on Tuesdays at 10 AM !**



Test 1

- Test 1 will include:
 - questions about the Syllabus
 - One question about the name of your instructor
 - Several questions about Markdown
 - 2 questions about the definition of Computational Thinking and the 6 skills (covered today)

This is **NOT** a
"GPA booster class"



A+ is rare in this course

Book your first Test!

<https://us.prairietest.com>

Learning Goals



Learning Goals

After this lecture, you should be able to:

- Explain the concept of Computational Thinking (CT)
 - Describe the relevance of CT in real-world contexts
- Identify the subskills that comprise CT
- Apply CT subskills to design and execute a structured solution
- Explain the key components of the Markdown language

Computational Thinking

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“Computational thinking is the thought processes involved in formulating problems and their solutions so that the solutions are in a form that can be effectively carried out by an information-processing agent”

[Cuny, Snyder, Wing]

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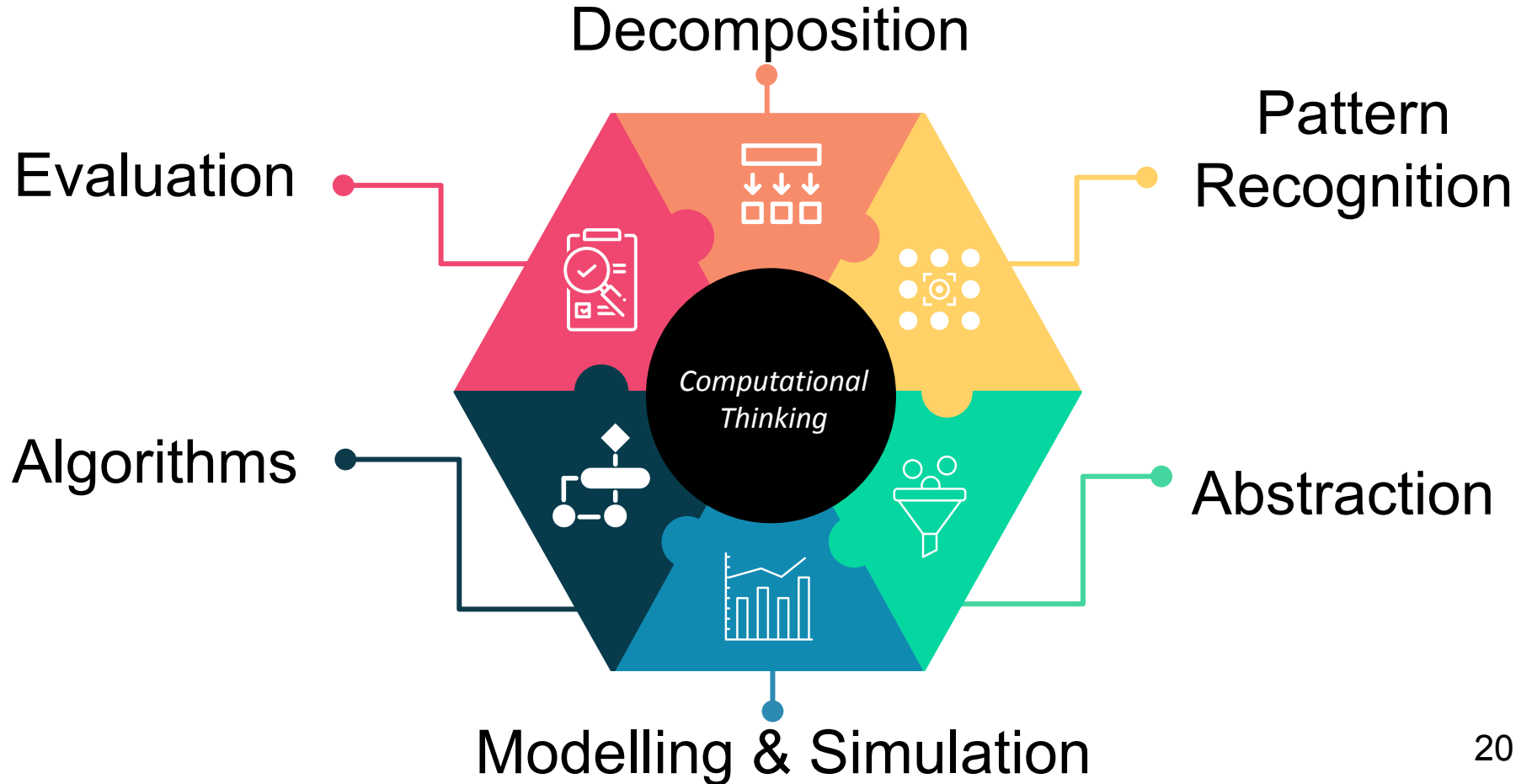
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Computational Thinking Skills





DECOMPOSITION

Breaking down problems into smaller, easier parts.



PATTERN RECOGNITION

Using patterns in information to solve problems.



ABSTRACTION

Finding information that is useful and taking away any information that is unhelpful.



MODELLING AND SIMULATION

Trying out different solutions or tracing the path of information to solve problems.



ALGORITHMS

Creating a set of instructions for solving a problem or completing a task.



EVALUATION

Assessing a solution to a problem and using that information again on new problems.

Class Activity



Class Activity: Sort the Cards

Imagine a robot must arrange a set of cards in ascending order (Ace to King, Same suit).

The robot can only follow your instructions.

Task [Groups of 3-4]

Create a clear set of steps/instructions to sort the cards





Robot Constraints

The robot has some limitations!

The robot can only:

- “**Recognize**” suits (♣, ♠, ♦, ♥) and these cards: 2-10, A, K, Q, J
- “**Look**” at one card at a time
- “**Pick up**” one card at a time
- “**Move**” one card to the beginning, middle,
- “**Remember**” the location and value of one previous card (not necessarily the last one it looked at)

Ed Discussion



Class Activity: Sort the Cards

1. Type the **final** steps your group came up with to sort the cards.
2. **Post your steps** on Ed Discussion.
3. **Choose** another group's steps, and try them to see if it works!





Wrap up



Wrap Up

- Computational Thinking (and card sorting)
- **Reminder:** Labs start next week!
- **Reminder:** Test 0!
- **Reminder:** Join Ed Discussion board (link on canvas)